

COMBINATORICS

NOTES JANUARY 15, 2026

Counting

“Those who do not count; do not count!”

Our combinatorial proofs will mainly rely upon the following principles

- (1) The Addition rule,
- (2) The Multiplication rule,
- (3) The Permutation rule (with or without repeating),
- (4) The Combination rule (with or without repeating),
- (5) The Distribution rule (twelve fold method),
- (6) Inclusion and Exclusion,
- (7) Generating Functions (ordinary and exponential),
- (8) Recurring Relations (linear and non-linear),
- (9) The Pigeonhole Principle (both simple and generalized) *important*,
- (10) Other miscellaneous methods.

Suppose we have boxes for distributing items We’d want to quickly ask some questions about



FIGURE 1. Three boxes

our boxes:

- are they labeled?
- are the items indistinguishable or have types?
- Define a bijection for capturing the mechanics of the problem.

Note. Maybe use the “Sterling” or “Burr” Numbers. To solve this problem we’d probably use rules 1, 2, 3, or 4 from above.

Before we get into the principles at hand let us note one definition.

Definition 1. A *partition* of a group of r identical objects divides the group into a collection of (unordered) subsets of various sizes. Analogously, we define a partition of the integer r to be a collection of positive integers whose sum is r . Normally we write this collection as a sum and list the integers in increasing order.

For example, the seven partions of the integer 5 are:

$$\begin{array}{ccccccc} 1 + 1 + 1 + 1 + 1 & 1 + 1 + 1 + 2 & 1 + 1 + 3 & & & & \\ & 1 + 2 + 2 & 1 + 4 & 2 + 3 & 5 & & \end{array}$$

Definition 2. *The Addition Principle.* Suppose a set S is partitioned into pairwise disjoint parts $S_1, S_2, \dots, S_m$, the number of the parts is given by adding the numbers obtained

$$|S| = \sum_{i=1}^m |s_i| = |s_1| + |s_2| + \dots + |s_m|$$

Consider the following example where we look at a situation where the subsets or parts are not necessarily disjoint

$$S = S_1 \cup S_2 \cup S_3 \cup S_4,$$

but we have

$$\begin{aligned} |S| &= |S_1| + |S_2| + |S_3| + |S_4| - |S_1 \cap S_2| - |S_1 \cap S_3| - |S_1 \cap S_4| \\ &\quad - |S_2 \cap S_3| - |S_2 \cap S_4| - |S_3 \cap S_4| \\ &\quad + |S_1 \cap S_2 \cap S_3| + |S_1 \cap S_2 \cap S_4| + |S_1 \cap S_3 \cap S_4| + |S_2 \cap S_3 \cap S_4| \\ &\quad - |S_1 \cap S_2 \cap S_3 \cap S_4| \end{aligned}$$

Definition 3. Let S be a st of ordered n -tuples $(a_1, a_2, \dots, a_n)$ of objects where the element a_j comes from a set of size n_j then the size of S is given by

$$|S| = \prod_{i=1}^n n_i$$

Example. Determine the nujber of integers that are the factors of $N = 3^4 \cdot 5^2 \cdot 11^7 \cdot 13^8$. We simply add one to the exponents and multiply them together giving us

$$5 \times 3 \times 8 \times 9 = 1080$$

Definition 4. *Permutations.* (with no repitition and order matters).

${}_nP_r$ is the permutation of n elements taking r at a time

$$n \times n - 1 \times n - 2 \times \dots \times n - r + 1 \times \frac{(n - r)(n - r - 1) \dots (1)}{(n - r)(n - r - 1) \dots (1)} = \frac{n!}{(n - r)!}$$

Example. Consider $\{a, b, c, d\}$ let's look at ${}_4P_3 = 4 \cdot 3 \cdot 2 = 24$

abc	bac	cab	dab
abd	bca	cad	dac
acd	bad	cba	dba
acb	bda	cbd	dbc
adb	bcd	cda	dca
adc	bdc	cdb	dcb

and if we drop the ordering then we have

$${}_nC_r = \frac{{}_nP_r}{r!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}.$$

Example. How many ways are there to rearrange the word *mississippi*? using $2!$ for the p 's, $4!$ for the s 's, $4!$ for the i 's and $1!$ for the m we count it as

$$\frac{11!}{2!4!4!2!}$$

In how many ways can we arrange the letters of *mississippi* so that the vowels are together? Well how about we just take all six of the i 's and block them together calling them I . Then, we are merely concerned with the word *mIsssspp*, and that counts out as:

$$\frac{8!}{4!2!}$$

Example. Bagel Shop. We have three types of bagels plain P , everything E , and cinnamon-rasin R . The combination for selecting 12 bagels follows as

$$P + E + R = 12$$

how may non-negative integer solutions does the equation have. For example the following are solutions

$$(6, 6, 0)$$

$$(4, 4, 4)$$

$$(3, 3, 6)$$

Also we can represent the bagles in the following pattern

$$\underbrace{bb}_P \mid \underbrace{bbbbbb}_E \mid \underbrace{bbbb}_R$$